Topic 3: Information Frictions and Nominal Rigidities

ECON 5322 — Macroeconomic Theory for Applications

Today's Topic

Lucas' Imperfect Information Model

- Policy implications: Sargent–Wallace, Barro

Outline:

1. The Lucas Imperfect Information Model
   - Only unanticipated money matters; signal extraction
2. Policy Implications: Sargent–Wallace on Policy Irrelevance
3. Time Series Tests (HW): Sargent on Observational Equivalence; Barro on Unanticipated Money
4. Extensions and Critiques
5. Nominal Rigidities: Contracting models (Fischer, Taylor)
6. Menu Costs (Mankiw, Blanchard–Kiyotaki)
7. Calvo (1983) and the NK Phillips Curve
8. New Keynesian Pricing Theory (Dixit–Stiglitz, Rotemberg)

Four Approaches to Aggregate Supply (1970s–90s)

Main questions:

- What determines the slope of the aggregate supply curve?
- Does money affect output? If so, what is the mechanism?

	Markets Clear: Yes	**Markets Clear: No**
Money Neutral: Yes	1. Classical / RBC Kydland & Prescott	3. Real Rigidity e.g. Efficiency wages (Akerlof, Yellen)
Money Neutral: No	2. Imperfect Information Friedman; Lucas '77	4. Nominal Rigidity Nominal contracts, menu costs (Fischer, Taylor, Calvo)

Emphasis throughout: micro-foundations + Rational Expectations

Imperfect Information Model — Motivation

Basic question: How to model "why money can affect output" in a RE setup?

- In the **Classical framework**: when $M\uparrow$, P and W go up, but Y stays fixed.
- **New Classical goal**: preserve frictionless markets *and* RE, yet show money can affect output (as empirically observed — Friedman–Schwartz, *A Monetary History of the US, 1867–1960*).
- **Tool**: Imperfect Information and Signal Extraction

Lucas Island Model (Imperfect information) — Setup

- J consumer/producers, each living on their own island, each producing good j
- **Supply** of good j depends on relative prices (all variables in logs):

Product j's price

$$y_{j,t} = \gamma\left[p_{j,t} - \mathbb{E}(P_t \mid \mathcal{I}_{j,t})\right]$$

Aggregate Price level

- **Information set** $\mathcal{I}_{j,t}$: at time t, producer j observes:
  - $p_{j,t}$ *and* all of its history (lags)
  - **Not** P_t (the overall price level at t) — only its history (lags)
  - So j must form the expectation $\mathbb{E}(P_t \mid \mathcal{I}_{j,t})$

Two types of shocks that affect $p_{j,t}$:

1. General monetary shock: $P_t \sim \mathcal{N}(\mu_t, \sigma^2)$
2. Sector-specific shock: $z_{j,t} \sim \mathcal{N}(0, \tau^2)$

$$p_{j,t} = P_t + z_{j,t}$$

$z \sim iid$

$\underbrace{0}$ $\quad \underbrace{Var(P) = \sigma^2}$

$Cov(p_j, P) = Cov(z, P) + Cov(P, P)$

$Var(p_j) = \sigma^2 + \tau^2$

J is large: individually, firms do not affect the aggregate P.

Signal Extraction

Q: Given observed $p_{j,t}$, how should producer j form $\mathbb{E}[P_t \mid \mathcal{I}_{j,t}]$?

Using the property of joint normal distributions: Var(P) Cov(P_t, $p_{j,t}$)

$$\begin{pmatrix} P_t \\ p_{j,t} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_t \\ \mu_t \end{pmatrix}, \begin{pmatrix} \sigma^2 & \sigma^2 \\ \sigma^2 & \underline{\sigma^2 + \tau^2} \end{pmatrix} \right)$$

Var($p_{j,t}$)

$$\underline{\mathbb{E}(P_t \mid p_{j,t})} = \mu_t + \frac{\sigma^2}{\sigma^2 + \tau^2}(p_{j,t} - \mu_t)$$

Substituting into the supply equation (and assuming $\underline{\mu_t = \mathbb{E}_{t-1} P_t = 0}$ for simplicity):

$$y_{j,t} = \gamma \left(\frac{\tau^2}{\tau^2 + \sigma^2} \right) p_{j,t}$$

- $\frac{\tau^2}{\tau^2+\sigma^2} < 1$ always: "signal extraction" **attenuates** output response
- If $\sigma^2 \to \infty$: all p_j movements read as inflation $\Rightarrow$ no need to adjust y_j
- If $\tau^2 \to \infty$: all movements are local $\Rightarrow$ y_j moves 1-for-1 with p_j
- Signal-to-noise ratio τ^2/σ^2: governs how much output reacts

Lucas Aggregate Supply Curve

Aggregate over all j producers (using joint-normal projection — see extra credit):

$$\boxed{Y_t = \gamma\left(\frac{\tau^2}{\tau^2+\sigma^2}\right)\left[P_t - \mathbb{E}_{t-1}P_t\right]}$$

Aggregate Supply

Key implications

- Only **unanticipated** movements in P affect output
- Anticipated monetary policy $\Rightarrow$ no output effect
- Slope of AS depends on τ^2/σ^2: the signal-to-noise ratio

Extra credit: Derive this formally using the joint-normal formula $\mathbb{E}(X_1|X_2) = \mu_1 + \frac{\mathrm{Cov}(X_1,X_2)}{\mathrm{Var}(X_2)}(X_2-\mu_2)$, with $X_1 = P_t$, $X_2 = p_{j,t}$, $\mu_2 = \mathbb{E}_{t-1}[P_t]$. Include the aggregation step from micro to macro variable.

Sargent–Wallace: Policy Irrelevance

To analyze policy, add Aggregate Demand to the Lucas AS:

$$Y_t = \gamma \left(\frac{\tau^2}{\tau^2 + \sigma^2} \right) \left[P_t - \mathbb{E}_{t-1} P_t \right]$$

Setup: Suppose at time t, the Fed observes all information up to $t-1$ (no timing advantage over the public). Let Q_{t-1} be a vector of observed variables:

$$m_t = b(L)\, Q_{t-1} + \varepsilon_t \qquad \text{(systematic rule plus non-systematic part)}$$

Since the public fully anticipates this rule, $\mathbb{E}_{t-1} P_t$ already incorporates it:

$$P_t - \mathbb{E}_{t-1} P_t = \varepsilon_t$$

Sargent–Wallace Result

Systematic monetary policy has no effect on output. The Fed can affect Y only by being erratic — i.e., by introducing genuine surprises.

Empirical Test: Lucas Cross-Regime Test

Model-implied hypothesis: Countries with more volatile monetary policy or AD shocks ($\sigma^2_{\Delta m}$) have a *steeper* AS: output reacts less & prices react more to AD shocks

Test: Estimate slope $\hat{\beta}_i$ from $\Delta y_i = \alpha + \beta_i \, \Delta m_i + \cdots$ for 18 countries.

Finding: Countries with larger $\sigma^2_{\Delta m}$ tend to have smaller $\hat{\beta}_i$ (steeper AS) — consistent with the model's prediction.

Caveats:

- **Correlation $\neq$ causation**: high-σ^2 countries also have high/volatile π
- Alternative mechanisms predict the same cross-country pattern:
  - High $\pi \Rightarrow$ fewer long-term nominal contracts $\Rightarrow$ less nominal rigidity $\Rightarrow$ steeper AS
  - High $\pi \Rightarrow$ firms willing to pay small menu costs $\Rightarrow$ more price flexibility
- Cannot identify whether imperfect information or nominal rigidity is the true cause

Critiques of the Imperfect Information Theory

1. **Correlation $\neq$ Causation (cross-regime test)**
   High $\sigma^2_{\Delta m}$ and small $\hat{\beta}$ shown, but causation not established; alternative mechanisms fit the same pattern.

2. **Implausibly large supply elasticity required**
   CPI announced monthly; changes typically $< 0.5\%$. To generate observed output fluctuations from such small signals requires an unrealistically large γ.

3. **Cost of announced disinflation**
   Under RE: a fully credible announced monetary contraction should not cause a recession. But Volcker's announced contraction (late 1970s) *did* cause a deep recession — inconsistent with the pure signal-extraction story.

4. **Is the information asymmetry plausible?**
   Optimizing agents have incentives to acquire cheap aggregate price information, which would erode the asymmetry.

Nominal Rigidities: Overview

Alternative view: Nominal rigidities create a role for monetary policy

- P does not adjust fast enough, so money has a temporary real effect

Lucas–Sargent–Wallace showed: if

1. Economy gravitates around a natural rate of output
2. Expectations are rational
3. Monetary authority follows a rule based on lagged variables

$\Rightarrow$ systematic monetary policy is irrelevant and undesirable (only surprise matters)

Fischer (1977) / Taylor (1980): These conditions are *necessary but not sufficient*.

- Rational Expectations does **not** imply policy irrelevance
- What additionally requires: **continuous market clearing via flexible prices**

Claim: If (nominal wage) contracts are "long enough," a deterministic monetary rule *can* still affect GDP and employment — even under RE.

$\Rightarrow$ MP can serve a **stabilization** purpose.

A Simple Contract Model (Not Fischer Yet)

System:

(1)	$w_t = \mathbb{E}_{t-1} p_t$	(wage-setting rule: 1-period stickiness)
(2)	$y_t = -(w_t - p_t)$	(AS)
(3)	$y_t + p_t = m_t + v_t, \quad v_t = d(L)v_{t-1} + \eta_t$	(AD)
(4)	$m_t = b(L)v_{t-1}$	(monetary policy rule)

Notes:

- $b_1, b_2, \ldots$ are policy choice parameters
- $\eta_t \sim \mathcal{N}(0, \sigma^2)$ is an iid AD shock

Without nominal friction (no wage rigidity): $w_t = p_t \Rightarrow y_t = 0$ at all times — AS is vertical; AD shocks cause no output fluctuations.

Solving the Simple Contract Model

Solution method: express endogenous variables in terms of exogenous ones.

Steps:

1. Sub (1) into (2): $$y_t = p_t - \mathbb{E}_{t-1} p_t \tag{5}$$
2. Sub (5) into (3): $$p_t = \tfrac{1}{2}(m_t + v_t) + \tfrac{1}{2}\mathbb{E}_{t-1} p_t \tag{6}$$
3. Take $\mathbb{E}_{t-1}$ of (6): $$\mathbb{E}_{t-1} p_t = \mathbb{E}_{t-1}\{m_t + v_t\} \tag{7}$$
4. Plug (7) and (6) into (5):

$$\boxed{y_t = \tfrac{1}{2}\,\eta_t}$$

result from HW1

Interpretations:

- **Sargent–Wallace holds**: all coefficients of $b(L)$ drop out — systematic MP is irrelevant, even with nominal rigidity (short contracts $\Rightarrow$ same result)
- Output is iid even though v_t is persistent: agents set wages at $t-1$ already knowing v_{t-1}, so they can predict v_t and p_t — only shock η_t surprises them

Fischer's 2-Period Staggered Contracts

- $\frac{1}{2}$ the population sets wages in *odd* periods ($t = 1, 3, 5, \ldots$);
  $\frac{1}{2}$ sets wages in *even* periods ($t = 2, 4, 6, \ldots$)
- Each wage contract lasts for **2 periods** (pre-determined pricing)

System:

Wage setting: (1) $w_t^{(i)} = \mathbb{E}_{t-i} p_t$, for $i = \{1, 2\}$

AS (2) $y_t = -\frac{1}{2}(w_t^{(1)} - p_t) - \frac{1}{2}(w_t^{(2)} - p_t)$

AD (3) $y_t + p_t = m_t + v_t, \quad v_t = D(L)v_{t-1} + \eta_t$

(4) $m_t = b(L)v_{t-1}$ → lag operator

→ Policy coefficients

To solve: substitute (1) into (2) ($\rightarrow$ (5)), then into (3)($\rightarrow$ (6)); take $\mathbb{E}_{t-1}$ and $\mathbb{E}_{t-2}$ to find $\mathbb{E}_{t-1} p_t$ and $\mathbb{E}_{t-2} p_t$; substitute back into AS (to get y_t).

Fischer: Solution for Output

After solving (see algebra Appendix):

Previous result w/o staggered contracts

$$\boxed{y_t = \frac{\eta_t}{2} + \frac{1}{3}(m_t - \mathbb{E}_{t-2} m_t) + \frac{1}{3} d_1 \eta_{t-1}}$$

Effect of policy

Key observation: Info from $t-1$ is unknown at $t-2$ when some contracts are set.

Compare two simple MP rules:

1. $m_t = \bar{m} = 0 \quad \Rightarrow y_t = \frac{1}{2}\eta_t + \frac{1}{3} d_1 \eta_{t-1}$, $\quad \mathrm{Var}(y_t) = \frac{1}{4}\sigma_\eta^2 + \frac{1}{9} d_1^2 \sigma_\eta^2$
2. $m_t = -d_1 \eta_{t-1} \quad \Rightarrow y_t = \frac{1}{2}\eta_t$, $\quad \mathrm{Var}(y_t) = \frac{1}{4}\sigma_\eta^2$

Rule (2) is **optimal**: residual variance comes only from shocks that neither policymaker nor agents can predict (Policy rule can be set to lower volatility)

$E_{t-2}(-d_1 \eta_{t-1}) = -d_1 \cdot \mathbb{E}_{t-2}(\eta_{t-1}) = 0 \Rightarrow \frac{1}{3}(m_t - \mathbb{E}_{t-2}(m_t)) = -\frac{1}{3} d_1 \eta_{t-1}$

Fischer: Why Does Systematic Policy Matter Now?

Three conditions for non-trivial policy relevance

1. **Information advantage (timing):** The Fed knows η_{t-1} when setting m_t, but this shock is *unknown to half of wage-setters* when they signed their contracts.
2. **Pre-formed contracts:** Sharing this info with the economy doesn't help — contracts are already in place. (Contrast with Lucas)
3. **Relevant information:** $d_1 \neq 0$, i.e., knowing η_{t-1} actually helps predict v_t. Otherwise, counter-cyclical MP is useless.

Conclusion: MP **can** stabilize output — a *Keynesian conclusion* from a RE model with nominal rigidity.

Indexation

Q: Why do agents write nominal wage contracts when only the *real* wage matters? Why not index wages to P?

In practice, there are both nominal *and* real shocks:

1. **Nominal shocks:** want to keep real wage fixed $\Rightarrow$ full indexation is optimal
2. **Real shocks:** want real wage to adjust to new marginal product of labor $\Rightarrow$ *no* indexation is optimal

With both types of shocks, the **optimal degree of indexation is intermediate**: some nominal rigidity is reasonable and rational. Justifying the Fischer mechanism.

Critiques of Contracting Models (1)

Are Fischer contracts rational?

Claim: No — stupid both ex ante and ex post.

- *Ex ante:* Workers and firms could be better off in a spot labor market. No clear benefit to locking in a long-term nominal contract.
- *Ex post:* Once agreed, both parties are stuck in an inefficient situation; both could gain from renegotiating once P_t is known.

Fischer's response:

- Contracts are observed empirically (often 3 years)
- Contracts may reflect convenient installment payment, not period-by-period wages for work; workload can vary within a contract
- Re-negotiation is costly; contracts may be optimal given a long-term relationship
- However: if contracted wage does not affect hiring period-to-period, its role in economic fluctuations is unclear

Critiques of Contracting Models (2)

Empirical real wage cyclicality:

In the Fischer model, the real wage is **counter-cyclical**:

$$\text{Positive AD shock} \Rightarrow P \uparrow \Rightarrow \frac{w}{P} \downarrow \Rightarrow L \uparrow \Rightarrow Y \uparrow$$

Problem: Data shows real wages are roughly **acyclical or slightly pro-cyclical** (Topic 1)

Also: monetary policy has **persistent effects beyond the contract length** in the data, whereas in Fischer's model, MP is effective only during the length of the contract.

$\Rightarrow$ Need additional mechanisms:
(a) productivity shocks [RBC],
(b) imperfect competition with countercyclical markups,
(c) pre-*fixed* staggered contracts + strategic complementarity [Taylor/Calvo].

Terminology: Pre-determined vs. Pre-fixed Contracts

	Fischer	**Taylor / Calvo**
Type	Pre-determined	Pre-fixed
Within contract	Prices can differ period-to-period	Price is the *same* throughout
2-period example	Set w_{t+1} and w_{t+2} separately	Set $w_{t+1} = w_{t+2}$
Based on	$\mathbb{E}_t P_{t+1}$ and $\mathbb{E}_t P_{t+2}$	$\mathbb{E}_t P_{t+1}$ and $\mathbb{E}_t P_{t+2}$

With fixed-length **pre-fixed** staggered contracts + **strategic complementarity**, shocks can propagate *beyond* the length of individual contracts (Taylor).

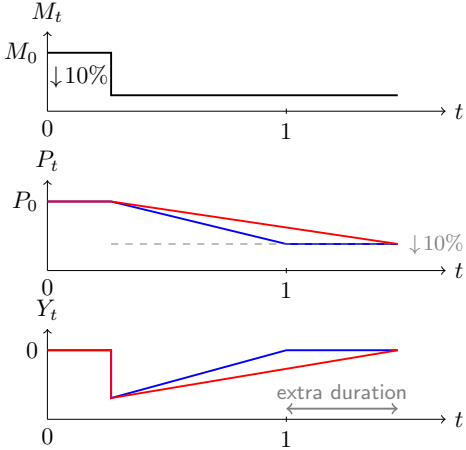
Taylor: Staggered Pre-Fixed vs. Strategic Complementarity

Staggered pre-fixed price (blue) vs. **Taylor with strategic complementarity** (red)

$p^*_{i,t} = M_t$ vs. $p^*_{i,t} = \theta M_t + \underbrace{(1-\theta)P_t}$

Strategic complementarity

(firms don't fully react to Mon. Policy because average aggregate prices are still high).



Empirics of Nominal Rigidities

Nature and behavior of nominal rigidity — a seminal empirical literature:

- **Substantial heterogeneity across sectors:**
  - Mean frequency of price change: ≈ 4.7 months; median: ≈ 11.5 months
  - Range: < 1 month (gasoline) to > 80 months (laundry machines)
- Many **sales / temporary price changes** that complicate measurement
- **Distribution of price changes matters:**
  - Large average size of changes
  - Large dispersion in size

Next: Why don't prices adjust/clear markets instantaneously?

Key Questions (EC)

- How does the real wage co-move with output in the Fischer model?
- What are the two main empirical failures of the Fischer staggered wage contract model?
- What elements or mechanisms can help address them and generate model outcomes more consistent with the data?

(Check Tech Slides 2: CES and Constrained Optimization refresher)

Menu Costs — Concept

Menu cost z: Any cost incurred directly in the process of (re-)setting prices

- Decision-making - managerial cost
- Physical repricing cost (e.g., printing menus)
- Near-rationality - "satisficing" (Akerlof & Yellen): close enough is good enough

Two stories for how menu costs may generate **monetary non-neutrality**:

1. **Partial Equilibrium (Mankiw):**
   Small (2nd-order) adjustment cost at the individual level can cause large (1st-order) aggregate social inefficiency

2. **General Equilibrium (Blanchard–Kiyotaki):**
   Aggregate demand externalities in a monopolistically competitive economy: price adjustment has a 1st-order social benefit but only a 2nd-order private benefit

Mankiw: Partial Equilibrium Setup

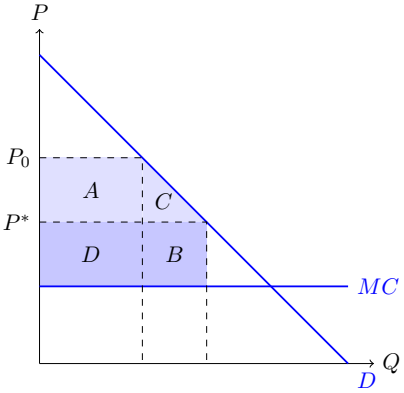
- $i = 1, \ldots, N$ producers, each with some monopoly power (monopolistic competition), each facing a downward-sloping demand curve
- At $t = 0$: firm sets $P = P_0$
- Unexpectedly, $M^s \downarrow$ (demand falls), so optimal price is $P^* < P_0$
- Firm must decide: incur menu cost z to adjust to P^*, or stay at P_0?

Intuition

Given menu cost z, even a small M^s shock can result in fixed P, implying output adjustment and **monetary non-neutrality**. The inefficiency arises from the externality on consumer surplus — firms ignore the social benefit of adjusting prices.

(Remember: RBC social planner = private markets outcome. Not here — externality breaks efficiency.)

Mankiw: Graphical Intuition



Firm's decision:

$$\begin{aligned}\Delta\pi &= \pi(P^*) - \pi(P_0)\\ &= (D+B) - (A+D)\\ &= B - A\end{aligned}$$

Changing price or not: $B - A \gtrless z$

$\Delta CS = A + C$

ΔSocial Surplus $= B + C$

$\Rightarrow$ If $B + C > z > B - A$:

Firms will **not** adjust price
$\Rightarrow$ Socially inefficient

Mankiw: Result

Decision rule:

- Firm adjusts if and only if: profit gain from adjusting $> z$
- Since the profit gain is 2nd order, even a *small* z can prevent adjustment

Social welfare:

- Society wants adjustment (1st-order welfare gain)
- Privately rational not to adjust $\Rightarrow$ **socially inefficient outcome**

Key claim:

- Firm's profit loss from staying at P_0: 2nd order (small for ϵ-deviations from P^*)
- Social welfare loss (including consumer surplus): 1st order

Taylor Exp of $\pi(P_0)$ around P^*

$$\pi(P_0) = \pi(P^*) + \underbrace{\pi'(P^*)}_{=0 \text{ (From FONC of profit maximization)}}(P_0 - P^*) + \frac{1}{2}\pi''(P^*)(P_0 - P^*)^2 + \dots$$

; $SW(P) = \pi(P) + CS(P)$

$$SW(P_0) = SW(P^*) + \underbrace{SW'(P^*)}_{\neq 0 \text{ in general}}(P_0 - P^*) + \frac{1}{2}SW''(P^*)(P_0 - P^*)^2 + \dots$$

Blanchard–Kiyotaki (AER 1987) — Setup

GE with aggregate demand externalities:

Continuum of household-producers on $[0, 1]$, indexed by i:

$$(1) \quad \text{Utility: } U_i = U\left(C_i, \frac{M_i}{P}, L_i\right)$$

$$(2) \quad \text{Production: } y_i = L_i$$

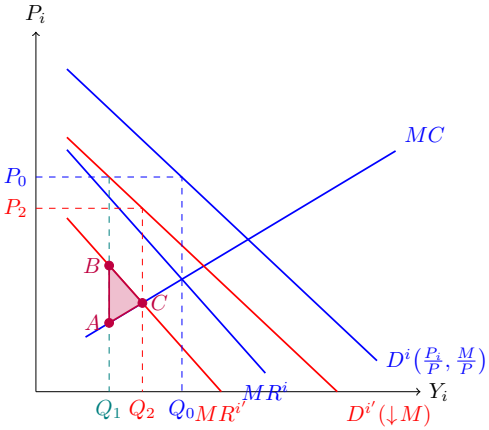
$$(3) \quad \text{Budget constraint: } p_i y_i = \int_0^1 p_j C_{ij} dj + M_i$$

- Dixit–Stiglitz CES aggregator; each producer has monopoly power over variety i

AD Externality: When firm i lowers its price:

1. Raises its own demand (movement along its demand curve) $\rightarrow$ *private benefit*
2. Lowers overall $P \Rightarrow$ raises real balances $M/P \Rightarrow \uparrow$ aggregate demand $\rightarrow$ *social benefit*
3. Most extra AD falls on *other* firms — firm i ignores this in its pricing decision
4. With menu cost: may not lower price to socially efficient level $\Rightarrow P$ too high, Y too low

BK: The Firm's Problem After $M^s \downarrow$



After $M^s \downarrow$:

- Demand shifts: $D^i \to D^{i'}$; $MR^i \to MR^{i'}$
- New optimum: $p_2 < p_0$, & $Q_2 < Q_0$,

Without menu cost:

- Firm moves to C: lowers price to P_2
- All firms adjust $\Rightarrow P \downarrow \Rightarrow M/P$ recovers $\Rightarrow$ output recovers

With menu cost:

- Firm stays at p_0, sells $Q_1 < Q_2$ (point A)
- Profit loss $\approx$ triangle ABC — 2nd order
- Social loss — 1st order

BK: Without Menu Cost

Negative money shock: $M^s \downarrow$

1. At Q_0: $MR > MC \Rightarrow$ firm raises Q_i, lowers p_i (to p_2)
2. Since firms are identical: all lower prices $\Rightarrow$ overall $P \downarrow$
3. M^s/P recovers $\Rightarrow D_i \uparrow \Rightarrow Q_i \uparrow = Y \uparrow$
4. **Output recovers fully.**

It is trivial to adjust prices: can reoptimize at no cost.

BK: With Menu Cost

If shock is small, profit loss from staying at p_0 may be $<$ menu cost z:

- Firms do not adjust $\Rightarrow$ real money balances and aggregate demand fall
- Output falls further below socially optimal level ($Q_0 \downarrow\downarrow$ to Q_1)

Results of the BK model

1. Monetary non-neutrality
2. **Pro-cyclical real wage** (unlike Fischer)
3. Justification for small monetary expansion: raise output toward social optimum
4. Large shocks: always socially costly (firms forced to pay menu costs)

Dynamics of Price Adjustment

Q: How to model the *dynamic* adjustment process under nominal rigidity in a micro-founded setup?

Approach	**Model**	**Micro-foundation for**
Time-contingent	Calvo (1983), Yun (1996)	Fischer / Taylor ideas
State-contingent	Rotemberg (1982)	Menu cost ideas
Others: Caplin–Spulberg (1997), Golosov–Lucas (2003)		

igskip **Empirical challenge:** Micro data show price changes are frequent and large; macro data show monetary policy effects are highly persistent. Need to reconcile both.

Calvo (1983) and Yun (1996)

Context: Micro-foundation for the New Keynesian Phillips Curve

- Firms adjust prices at **random intervals**, staggered fashion
- Opportunity to adjust arrives as a Poisson process: constant probability $1-\theta$ of adjusting at each t
- Expected duration between adjustments: $\dfrac{1}{1-\theta}$
- Empirics / calibration: mean frequency ≈ 4.7 months $\Rightarrow 1-\theta \approx 0.21$, $\theta \approx 0.79$
- Time-contingent with *exogenous* probability of price change

See handout for step-by-step derivations. Also: Romer 7.4; Walsh Ch.6 p.249.

Calvo: Firm Optimization

Per-period quadratic loss for a firm setting price at x_t:

$$\ell(x_t, p_t^*) = \frac{\kappa}{2}(x_t - p_t^*)^2$$

where x_t = price chosen; p_t^* = frictionless optimal price at t.

Minimize expected discounted loss:

$$\min_{x_t} \sum_{j=0}^{\infty} (1-\theta)(\beta\theta)^j \frac{\kappa}{2} \left[(\mathbb{E}_s p_{t+j}^* - x_t)^2 \right]$$

Optimal reset price: $x_t = (1-\beta\theta) \sum_{k=0}^{\infty} (\beta\theta)^j \, \mathbb{E}_s[p_{t+j}^*]$

Forward-looking weighted average of expected future optimal prices, discounted by $\beta\theta$.

Aggregate over all firms + **first-difference** $\Rightarrow$ New Keynesian Phillips Curve:

$$\boxed{\pi_t = \frac{(1-\theta)(1-\beta\theta)}{\theta} \phi y_t + \beta \, \mathbb{E}_t[\pi_{t+1}]},$$

New Keynesian Theory: Monopoly Pricing

Supply-side price setting and towards a micro-founded menu costs model

Relevant Market Structures

RBC / Neoclassical:

- All prices determined in perfect competition; firms are *price takers*
- No firm has market power; private equilibrium is efficient

New Keynesian:

- Firms *set* their own nominal prices — they have market power
- **Markup** $\mu = P/MC \geq 1$: price relative to marginal cost
  - Perfect competition: $\mu = 1$; Monopolistic competition: $\mu > 1$
- Re-setting prices may entail costs ("menu costs", interpreted broadly)
- **Central question:** How do sticky prices affect monetary policy?

Upcoming analysis:

Step 1 Develop theory in which firms are purposeful price setters (Dixit–Stiglitz)

Step 2 Superimpose costs of re-setting nominal prices (Rotemberg)

Monopolistic Competition

Monopolistically-competitive goods markets: the foundation of NK theory.
An intermediate structure between perfect competition and pure monopoly.

Key features:

- Allows purposeful price setting; retains some competitive features
- Goods are **imperfect substitutes** — the essence of market power
- **Markup** $\mu = P/MC$: key concept in monopoly/MC theory
- Constrained output: justifies policy intervention (market no longer efficient)
- Firms willing to sell more since $P > MC$; output is demand-determined

NK model operationalizes this by dividing goods markets into two sectors: retail (competitive) and wholesale (monopolistically competitive).

NK Model Overview: Two-Sector Structure

Retail firms:

- Sell a perfectly-substitutable retail good in a competitive market
- Purchase differentiated wholesale goods; bundle them (Dixit–Stiglitz aggregator)
- No labor or other inputs — like a retail store that produces nothing of its own

Wholesale firms:

- Infinite continuum; each produces an imperfectly substitutable variety
- Imperfect substitutes: a monopolist of its variety, but competes with all others
- Each wholesale firm is a **price setter**

Why this structure?

- Allows separate analysis of micro-level pricing and macro-level aggregation
- Retail problem yields the demand function each wholesale firm faces
- Wholesale problem pins down equilibrium pricing and the NKPC

Retail Firms: Dixit–Stiglitz Aggregator

Dixit–Stiglitz "production function":

$$y_t = \left[\int_0^1 y_{it}^{1/\varepsilon}\, di\right]^{\varepsilon}$$

- $\varepsilon > 1$: curvature. $\frac{\varepsilon}{\varepsilon-1}$: elast. of substitution between any pair of wholesale goods
- ES$\rightarrow \infty$: perfect substitutes $\Rightarrow$ recovers perfect competition (RBC)
- ε also the key determinant of the profit-maximizing markup
- Workhorse of NK theory; also central in international trade theory

Retail firm's profit function:

$$P_t y_t - \int_0^1 P_{it}\, y_{it}\, di$$

$$= P_t \left[\int_0^1 y_{it}^{1/\varepsilon} di\right]^{\varepsilon} - \int_0^1 P_{it}\, y_{it}\, di$$

Retail Firms: Demand for Wholesale Good j

Retail firm's problem:

$$\max_{\{y_{it}\}} \; P_t \left[\int_0^1 y_{it}^{1/\varepsilon} di \right]^{\varepsilon} - \int_0^1 P_{it}\, y_{it}\, di$$

Chooses profit-maximizing quantity of each wholesale input. Focus on any good j:

FOC with respect to y_{jt}:

$$\vdots$$

(after several rearrangements)

$$\boxed{y_{jt} = \left(\frac{P_{jt}}{P_t} \right)^{\frac{\varepsilon}{1-\varepsilon}} y_t}$$

(Demand function for wholesale good j)

Demand is elastic with elasticity $\frac{\varepsilon}{\varepsilon - 1}$; depends only on the *relative price* P_{jt}/P_t.

Wholesale Firms: Setup

Focus on arbitrary wholesale firm j. By symmetry: all firms make analogous decisions

Production assumptions:

- Zero fixed costs
- Constant returns to scale (CRS): scaling all inputs by x scales output by x
- No need to specify factors of production or functional form of $f(\cdot)$
- Marginal cost = average cost; mc **is NOT a function of quantity**

Profit function (period t):

$$P_{jt}\, y_{jt} - P_t \cdot mc_{jt} \cdot y_{jt}$$

- Revenue in terms of firm's own price P_{jt}
- Production costs converted to nominal using P_t (not P_{jt}): costs not denominated in firm's own price

Wholesale Firms: Profit Maximization

Substitute in the demand function $y_{jt} = (P_{jt}/P_t)^{\frac{\varepsilon}{\varepsilon-1}} y_t$:

$$\max_{P_{jt}} \; P_{jt}\left(\frac{P_{jt}}{P_t}\right)^{\frac{\varepsilon}{\varepsilon-1}} y_t \; - \; P_t \cdot mc_{jt} \cdot \left(\frac{P_{jt}}{P_t}\right)^{\frac{\varepsilon}{\varepsilon-1}} y_t$$

The firm internalizes the effect of its price on quantity sold.

FOC with respect to P_{jt} (after rearrangement):

$$\frac{1}{1-\varepsilon} P_{jt}^{\frac{\varepsilon}{1-\varepsilon}} P_t^{\frac{\varepsilon}{\varepsilon-1}} y_t + \frac{\varepsilon}{1-\varepsilon} P_{jt}^{\frac{2\varepsilon-1}{1-\varepsilon}} P_t^{\frac{2\varepsilon-1}{\varepsilon-1}} mc_{jt} y_t = 0$$

Solving:

$$\boxed{\frac{P_{jt}}{P_t} = \varepsilon \cdot mc_{jt} \longrightarrow p_{jt} = \varepsilon \cdot mc_{jt} > mc_{jt}}$$

Optimal relative price is a **constant markup** $\varepsilon > 1$ over marginal cost:
$\mu_{jt} = \frac{p_{jt}}{mc} = \varepsilon$ Key result of Dixit–Stiglitz theory.

The Dixit–Stiglitz Framework: Implications

Key prediction: Firms aim to maintain prices at a constant markup over marginal cost:

$$\mu_{jt} = \frac{P_{jt}}{P_t \cdot mc_{jt}} = \frac{p_{jt}}{mc_{jt}} = \varepsilon = \text{const.}$$

Empirical relevance?

- Not very good in the short run
- Perhaps better in the long run
- Markups generally **countercyclical** at business-cycle frequencies: expansions $\Rightarrow$ markups fall; recessions $\Rightarrow$ markups rise

Applications of DS framework: customer switching, brand loyalty, search costs.

Next: Use Dixit–Stiglitz as the foundation for New Keynesian sticky-price models.

New Keynesian Theory: The Modern Sticky-Price Model

Rotemberg (1982) — Menu Costs Model

Rotemberg: Relevant Market Structures (Recap)

RBC: No firm is a price setter. Prices determined in perfect competition.

New Keynesian:

- Firms set nominal prices; some market power
- Re-setting prices entails costs (menu costs, interpreted broadly)

Steps:

Step 1 (done) Dixit–Stiglitz: theory of purposeful price-setting

Step 2 (now) Superimpose costs of re-setting nominal prices $\Rightarrow$ sticky prices

Menu Costs: Fixed vs. Convex

Two views of the nature of menu costs:

1. **Fixed menu cost:** total cost is independent of the *magnitude* of the price change
   - Example: printing new restaurant menus — cost is the same whether price changes by 5% or 50%
2. **Convex menu cost:** cost increasing and convex in the magnitude of the change
   - Example: "cost of angering customers," managerial time, renegotiation costs — plausibly increase with larger adjustments

Both are likely aspects of real-world menu costs.

Formal NK theory (Rotemberg): focuses on **convex** menu costs — tractable and yields a smooth Phillips curve.

(Anderson–Simester; Zbracki et al. provide empirical evidence on menu cost magnitudes)

Rotemberg: Modeling Convex Menu Costs

Menu costs introduced at the **wholesale firm level** (they are the price setters).

Wholesale firm j incurs a real convex menu cost of nominal price adjustment:

$$\frac{\psi}{2}\left(\frac{P_{jt}}{P_{j,t-1}}-1\right)^2 y_t$$

- $\psi > 0$: governs importance of menu costs; $\psi = 0$ recovers basic DS
- **Real cost** denominated in goods; requires factor P_t for nominal conversion
- Convex: larger $\%$ deviation of P_{jt} from $P_{j,t-1} \Rightarrow$ larger menu cost
- Disincentive to adjusting prices "too quickly"

Key implication: The pricing problem is now **dynamic** — price chosen today affects menu cost penalty next period.

Q: Are downward adjustments as costly as upward ones? Intuition says no. Anderson–Simester evidence: maybe.

Retail Firms: Unchanged from Basic DS

With menu costs, the retail firm's problem is unchanged:

$$\max_{\{y_{it}\}} P_t \left[\int_0^1 y_{it}^{1/\varepsilon} di \right]^{\varepsilon} - \int_0^1 P_{it}\, y_{it}\, di$$

FOC with respect to y_{jt}:

$$\vdots$$

$$y_{jt} = \left(\frac{P_{jt}}{P_t} \right)^{-\varepsilon} y_t$$

IDENTICAL to flexible-price DS framework

Menu costs are a wholesale-firm phenomenon; the demand side is unaffected.

Wholesale Firms: Dynamic Problem

Wholesale firm j now also incurs quadratic menu costs. Period-t profit:

$$P_{jt}\, y_{jt} - P_t \cdot mc_{jt} \cdot y_{jt} - \frac{\psi}{2}\left(\frac{P_{jt}}{P_{j,t-1}} - 1\right)^2 P_t$$

Menu cost in period $t+1$ depends on P_{jt} (see denominator) $\Rightarrow$ **dynamic problem**.

Two-period forward-looking profit function:

$$\max_{P_{jt}} P_{jt}\, y_{jt} - P_t\, mc_{jt}\, y_{jt} - \frac{\psi}{2}\left(\frac{P_{jt}}{P_{j,t-1}} - 1\right)^2 P_t$$
$$+ \frac{\beta}{1+\pi_{t+1}}\left[P_{j,t+1}\, y_{j,t+1} - P_{t+1}\, mc_{j,t+1}\, y_{j,t+1} - \frac{\psi}{2}\left(\frac{P_{j,t+1}}{P_{jt}} - 1\right)^2 P_{t+1}\right]$$

s.t. retailer (optimal) demand.

- Discount factor adjusted for inflation: $\beta/(1+\pi_{t+1})$

Wholesale Firms: Substituting Demand and Taking FOC

Substitute demand $y_{jt} = \left(P_{jt}/P_t\right)^{-\varepsilon} y_t$ into both periods:

$$\max_{P_{jt}} \; P_{jt}\left(\frac{P_{jt}}{P_t}\right)^{\frac{\varepsilon}{\varepsilon-1}} y_t - P_t\, mc_{jt}\left(\frac{P_{jt}}{P_t}\right)^{\frac{\varepsilon}{\varepsilon-1}} y_t - \frac{\psi}{2}\left(\frac{P_{jt}}{P_{j,t-1}} - 1\right)^2 P_t$$
$$+\frac{\beta}{1+\pi_{t+1}}\left[P_{j,t+1}\left(\frac{P_{j,t+1}}{P_{t+1}}\right)^{\frac{\varepsilon}{\varepsilon-1}} y_{t+1} - P_{t+1}\, mc_{j,t+1}\left(\frac{P_{j,t+1}}{P_{t+1}}\right)^{\frac{\varepsilon}{\varepsilon-1}} y_{t+1} - \frac{\psi}{2}\left(\frac{P_{j,t+1}}{P_{jt}} - 1\right)^2 P_{t+1}\right]$$

The firm chooses P_{jt} and internalizes its effect on:

- Current revenue and cost via demand (as in static DS)
- Current menu cost (deviation from $P_{j,t-1}$)
- *Future* menu cost next period (deviation of $P_{j,t+1}$ from P_{jt})

$\Rightarrow$ **Take FOC with respect to** P_{jt}:

Wholesale Firms: FOC

FOC with respect to P_{jt}:

$$\frac{1}{1-\varepsilon} P_{jt}^{\frac{\varepsilon}{1-\varepsilon}} P_t^{\frac{\varepsilon}{\varepsilon-1}} y_t - \frac{\varepsilon}{1-\varepsilon} P_{jt}^{\frac{2\varepsilon-1}{1-\varepsilon}} P_t^{\frac{2\varepsilon-1}{\varepsilon-1}} mc_{jt} y_t - \psi \left(\frac{P_{jt}}{P_{jt-1}} - 1 \right) \frac{P_t}{P_{jt-1}} + \frac{\beta\psi}{1+\pi_{t+1}} \left(\frac{P_{jt+1}}{P_{jt}} - 1 \right) \frac{P_{t+1}}{P_{jt}} \frac{P_{jt+1}}{P_{jt}} = 0$$

Three forces in the FOC:

1. Standard DS markup motive: first two terms (MR and MC effects) — together drive price toward the optimal markup $\varepsilon \cdot mc$
2. Current menu cost: third term — penalizes deviation of P_{jt} from $P_{j,t-1}$
3. Future menu cost: fourth term — forward-looking benefit of adjusting today: a larger P_{jt} reduces next period's adjustment cost

When $\psi = 0$: terms 3 and 4 vanish $\Rightarrow$ collapses to the static DS rule $P_{jt}/P_t = \varepsilon\, mc_t$.

Symmetric Equilibrium

Drop the j distinction: impose $P_{jt} = P_t$ for all j (symmetric equilibrium).

Use $\pi_t = P_t/P_{t-1} - 1$. Several terms simplify (e.g., $P_{jt}/P_t = 1$):

The FOC becomes:

$$\frac{1}{1-\varepsilon}(1-\varepsilon\, mc_t)y_t - \psi\left(\frac{P_t}{P_{t-1}} - 1\right)\frac{P_t}{P_{t-1}} + \frac{\beta\psi}{1+\pi_{t+1}}\left(\frac{P_{t+1}}{P_t} - 1\right)\frac{P_{t+1}}{P_t}\frac{P_{t+1}}{P_t} = 0$$

Several terms combine using $P_t/P_{t-1} = 1 + \pi_t$ and analogous expressions, yielding:

$$\frac{1}{1-\varepsilon}(1-\varepsilon\, mc_t)y_t - \psi\,\pi_t(1+\pi_t) + \frac{\beta\psi}{1+\pi_{t+1}}\,\pi_{t+1}(1+\pi_{t+1})^2 = 0$$

Exact (non-linear) **New Keynesian Phillips Curve** from the Rotemberg model.

New Keynesian Phillips Curve (Rotemberg) - log-linearized

Log-linearize the NKPC:

$$\frac{1}{1-\varepsilon}\bar{y}\hat{y}_t - \frac{\varepsilon}{1-\varepsilon}\overline{mc}\,\bar{y}\,(\widehat{mc}_t + \hat{y}_t) - \psi\bar{\pi}\hat{\pi}_t - \psi\bar{\pi}^2 2\hat{\pi}_t + \beta\psi\bar{\pi}\hat{\pi}_{t+1} + \beta\psi\bar{\pi}^2 2\hat{\pi}_{t+1} = 0$$

$$\boxed{\hat{\pi}_t = \gamma_1\hat{y}_t - \gamma_2\,\widehat{mc}_t + \beta\,\mathbb{E}_t[\hat{\pi}_{t+1}]}$$

- Links period-t inflation to period-t marginal cost and $(t+1)$ expected inflation
- **Forward-looking:** pricing decisions are inherently dynamic
- Higher ψ (stronger menu costs) $\Rightarrow$ flatter NKPC (more sluggish adjustment)

Comparison with Calvo–Yun: Use that $\hat{mc}_t \propto \hat{y}_t$

$$\pi_t = \underbrace{\lambda}_{\kappa_{\text{Rot}}}\widehat{y}_t + \beta\,\mathbb{E}_t[\pi_{t+1}] \qquad \text{vs.} \qquad \pi_t = \underbrace{\frac{(1-\theta)(1-\beta\theta)}{\theta}}_{\kappa_{\text{Calvo}}}\widehat{y}_t + \beta\,\mathbb{E}_t[\pi_{t+1}]$$

Identical structure: same reduced form, different structural interpretation of the slope

Wrapping Up: Aggregate Supply

What we have covered:

- Rational Expectations revolution and micro-foundation approach
- How to model a sloped AS curve:
  - *New Classical (Lucas):* imperfect information $\Rightarrow$ only surprises matter; systematic MP irrelevant
  - *New Keynesian (Fischer, Taylor, Calvo):* nominal rigidities $\Rightarrow$ systematic MP can stabilize
- Empirical support and critiques of each approach
- Menu costs and AD externalities: micro-foundations for non-neutrality
- New Neoclassical Synthesis: DSGE with RE + monopolistic competition + nominal rigidities $\Rightarrow$ NKPC as the AS equation

Are we done? What are the remaining building blocks of the model?

Next: Consumption and Aggregate Demand

(See Tech Slides 2: CES functions and constrained optimization refresher)

Appendix

Appendix 1: Fischer Solution — Algebra Steps (1/3)

Substitute (5) into (3) and solve for p_t:

$$p_t = \frac{1}{2}m_t + \frac{1}{2}v_t + \frac{1}{4}E_{t-1}p_t + \frac{1}{4}E_{t-2}p_t$$

Taking expectations:

$$E_{t-1}p_t = \frac{2}{3}E_{t-1}m_t + \frac{2}{3}E_{t-1}v_t + \frac{1}{3}E_{t-2}p_t$$

$$E_{t-2}p_t = E_{t-2}m_t + E_{t-2}v_t$$

Substitute $E_{t-2}p_t$ into the expression for $E_{t-1}p_t$:

$$E_{t-1}p_t = \frac{2}{3}E_{t-1}m_t + \frac{1}{3}E_{t-2}m_t + \frac{2}{3}E_{t-1}v_t + \frac{1}{3}E_{t-2}v_t$$

Appendix 1: Fischer Solution — Algebra Steps (2/3)

Replace both expectations in the equation for p_t:

$$p_t = \frac{1}{2}m_t + \frac{1}{6}E_{t-1}m_t + \frac{1}{3}E_{t-2}m_t + \frac{1}{2}v_t + \frac{1}{6}E_{t-1}v_t + \frac{1}{3}E_{t-2}v_t$$

Find expressions for $E_{t-1}p_t - p_t$ and $E_{t-2}p_t - p_t$ and substitute into AS:

$$E_{t-1}p_t - p_t = -\frac{1}{2}\eta_t$$

$$E_{t-2}p_t - p_t = \frac{2}{3}E_{t-2}m_t + \frac{2}{3}E_{t-2}v_t - \frac{4}{6}m_t - \frac{4}{6}E_{t-1}v_t - \frac{1}{2}\eta_t$$

Substituting into AS gives:

$$y_t = \frac{\eta_t}{2} + \frac{1}{3}(m_t - E_{t-2}m_t) + \frac{1}{3}(E_{t-1}v_t - E_{t-2}v_t)$$

Appendix 1: Fischer Solution — Algebra Steps (3/3)

Finally, use that $v_t = d_1 v_{t-1} + d_2 v_{t-2} + \cdots$:

$$\begin{aligned} E_{t-1}v_t - E_{t-2}v_t &= E_{t-1}(d_1 v_{t-1} + d_2 v_{t-2} + \cdots) - E_{t-2}(d_1 v_{t-1} + d_2 v_{t-2} + \cdots) \\ &= E_{t-2}d_1 v_{t-1} + d_1 \eta_{t-1} - E_{t-2}d_1 v_{t-1} \\ &= d_1 \eta_{t-1} \end{aligned}$$

Substituting back:

$$\boxed{y_t = \frac{\eta_t}{2} + \frac{1}{3}(m_t - E_{t-2}m_t) + \frac{1}{3}\, d_1 \eta_{t-1}}$$